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Design and Implementation of a Star-Tracker for LEO Satellite

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Abstract

An important task of the attitude determination and control subsystem (ADCS) is the determination of attitude of the satellite in reference to the Earth or any other planetary and correct any deviation from the expected attitude. In order to determine the attitude of the satellite, sensors are needed to sense the orientation of the satellite. The star-tracker is able to determine the attitude from the view field of the star set without any prior knowledge of sensor information. In this paper, the design, simulation and implementation of a star-tracker is presented for LEO satellites. For capturing image data, the MT9P031 CMOS image sensor from Aptina Imaging was used. The star tracker uses a Texas Instrument OMAP3530 using an ARM Cortex A8. The processor supports powerful graphic interfaces resulting in reduction of the size and consumption power of the star-tracker. For storage and processing, the star tracker used NAND flash memory. The communication interface to the satellite is done through CAN Bus. One of the key features of this sensor is processing various algorithms for image processing, positioning, and tracking according to mission requirements. Quaternion estimator (QUEST) algorithm was utilized for positioning. The tracking algorithm utilizes information from the previous frame of the star-tracking. This algorithm is a high-speed, low cost, and low memory implementation. In the field tests, the identification success rate of this system was 96.81%.

Keywords: star tracker, attitude determination, Image processing, tracking.

Introduction

To succeed in a space mission, it is vital to have a reliable attitude determination system. To determine and control the attitude, spacecrafts normally use several sensors, such as magnetic sensors and earth sensor, that work in a closed loop with actuators or torquers for the correction of the orientation of the spacecraft. It is also possible that the horizon sensor provides useful information when it is close to any large celestial object [1]. Nevertheless, these measuring devices are not applicable when the spacecraft moves from one planet to another. Some other measuring devices can provide attitude information for the spacecraft control system during long travels. Equipment such as gyroscopes, sun sensors, and star-trackers are generally used onboard for attitude determination tasks. Each of these sensors has its own advantages and disadvantages. Gyroscopes provide continuous and instantaneous information relative to a past

state. Therefore, in the case that the current state is lost, gyroscopes fail to work. This may be lost due to a failure in the satellite energy subsystem or resetting the sensor by the onboard computer.

Sun sensors are inexpensive and lightweight, but require the movement of the sensor (additional mechanical mechanism) or possibly the entire spacecraft to find the sun. Additional mechanism means increased hardware, weight and cost. Moreover, the additional mechanical mechanism reduces the total reliability of the sensor. However, it is possible to use multiple onboard sun sensors, one of each panel of the spacecraft, that attitude can be determined based on information from all sensors, without the need of spacecraft rotation. In this scenario, the weight and power consumption of the sensor increases several times. The star trackers are the most accurate and allow attitude determination without previous information or rotation of the sensor.

In star sensors, stars are extracted from a current image taken by the sensor. Then, the detected stars are compared to the stored star catalog to determine which stars are in the sensor field of view. Then further images processing is done to enable the star sensor to determine the stars observed and the spacecraft's attitude. Therefore, selection of the star catalog is an important task, as it gives the framework for a good functioning star identification [2]. In our star sensor, the Hipparcos star catalog will be used. It was shown that the position of a rigid object could be completely described by considering a set of Cartesian coordinate systems, which fixed to the object [3, 4]. In this sensor, attitude determination is performed based on the quaternion parameters. The quaternion estimation (Quest) algorithm is chosen for attitude determination. In the first generation of star tracker sensors, a few bright stars were identified [5] and their centroid was given as an output to the spacecraft processor, so no attitude information was directly provided. Star identification and consequently attitude determination should have been performed outside of the star sensors. Therefore, the recorded image was transmitted to the ground segment, and attitude determination was performed by the ground station after receiving the star image. Due to the limited onboard resources as well as high complexity of the star identification algorithms (which require comparing the stars in an image and the reference catalog), attitude determination had not been implemented onboard by the late seventies. In the last decade, with the availability of powerful microcomputers, the comparison of the observed stars with the stars of the guide star catalog in the satellite onboard became possible. As a result, the rotation of a reference coordinate system relative to the star-tracking coordinate system can be calculated directly and onboard. Especially, the rapid development of imaging and digital algorithms have provided high reliable positioning algorithms with fast updating rate.

Vagle, Soosaar, and Iuzzalino first introduced utilizing star-tracker for attitude determination in 1969 [6]. Some well-known sensors are as follows: SHEFEX 2 star tracker includes a suitable charge-coupled device (CCD) camera with a corresponding lens and a PC/104 computer for the image processing chain [7]. It used the monochromatic CCD camera (model:

PROSILICA EC655). ASTRO star tracker family was continuously improved and enlarged with the models ASTRO5, ASTRO10 and ASTRO15. All of them are CCD detector based [8]. Today ASTRO 15 is the most sold star tracker worldwide and is operating on several geotelecom as well as Earth observation satellites. The autonomous star sensor ASTRO APS has been qualified in 2008/09 [9] for a wide range of space application environments and is already scheduled for a variety of Earth observation and geo-stationary telecommunication programs. The ASTRO APS star sensor is characterized by the application of the Complementary Metal Oxide Semiconductor (CMOS) active pixel detector technology and the hardware/software implementation of new and advanced measures in autonomous operation and functionality. Authors in [10] introduce a digital imaging system, in which the previous master controller field-programmable gate array (FPGA) was replaced by ARM+DSP (digital signal processor) stronger in processing and operation, and CMOS imaging sensor and ARM+DSP are the core components of a whole circuit. In this system, TMS320DM3730 was used as the main control unit; the MT9P031 made by Aptina was selected to collect images. The CubeSat star tracker contains Melexis image sensor (MLX75412). An FPGA and an SRAM module which together act as a frame buffer [11]. Zunzong Yea et al. designed and implemented a Star Sensor Imaging Driver Module based on IMX224. They chose a FPGA that embedded Cortex-M3 ARM soft core as imaging driver chip [12].

The organization of this paper is as follows: In section 2, the principles of the star tracker and processing algorithms will be expressed. In section 3, the sensor hardware will be expressed. In section 4, the test results will be presented. Finally, Section 5 concludes the article.

2- Functional principles and algorithms used in the star tracker

Figure 1 shows the block diagram of a typical star tracker. After receiving the star image from the star tracker camera, the following steps and processes are performed in the star sensor processor:

- Image processing and star centroid determination
- Using matching algorithms to identify stars in the image
- Extract satellite position

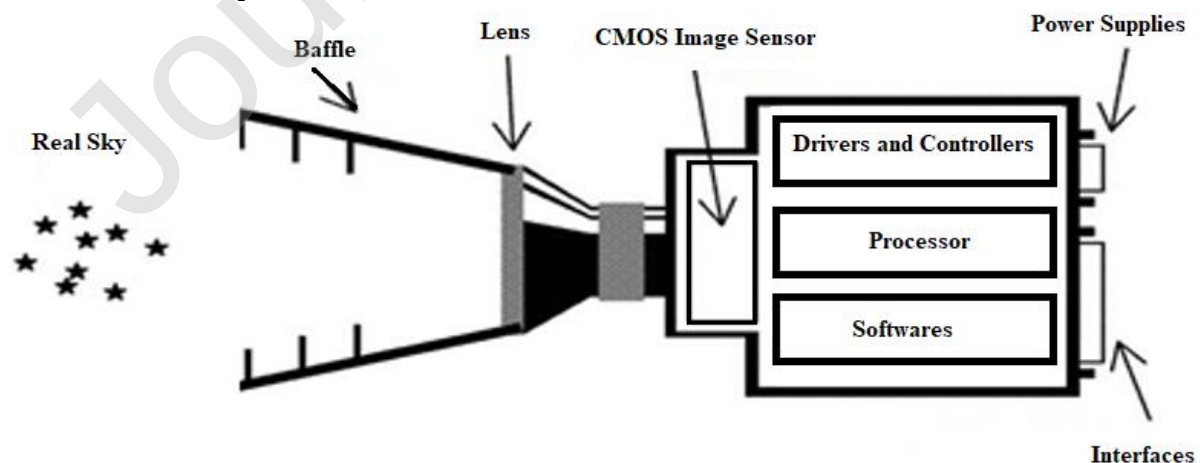


Fig. 1. Block diagram of a typical star tracker [13]

Figure 2 shows the attitude determination processes of a star tracker [14]. At first, the image taken from the star tracker camera is processed to detect the stars in it. The light points that are within the range defined for the star tracker are detached from the image, and then centroid of the existing stars is extracted. After that, using the selected algorithms, a given feature is created for the identified stars. For example, if the triangle method is selected as the matching base, required triangles are created using the stars specified in the image. In the next step, by searching the database, the matching between the captured stars and the stars in the catalog is performed. Then, Euler or quaternion transforms are extracted to match the orientation of the captured stars with the orientation of those of the star catalog. These transformation matrices are in fact illustrating the satellite position.

The state diagram of the star tracker is shown in Fig. 3. As shown, the star tracker, after initial setup, goes to the initial recovery mode. In this mode, by imaging and using star catalog, star-tracker recognizes its attitude quaternion. After the initial diagnosis of the position, the star tracker switches to tracking mode. In this mode, using a more limited image frame and a small part of the catalog, attitude quaternion is updated. If the attitude information is lost during the tracking mode, the star-tracker returns to the initial recovery mode and continues as mentioned. In each step of the operation, if any error is detected in the system, the star tracker returns to the failure mode and returns to the initialize mode after failure recovery. In the other word, In case of attitude determination failure, the star tracker goes back to the lost in space mode (LOS).

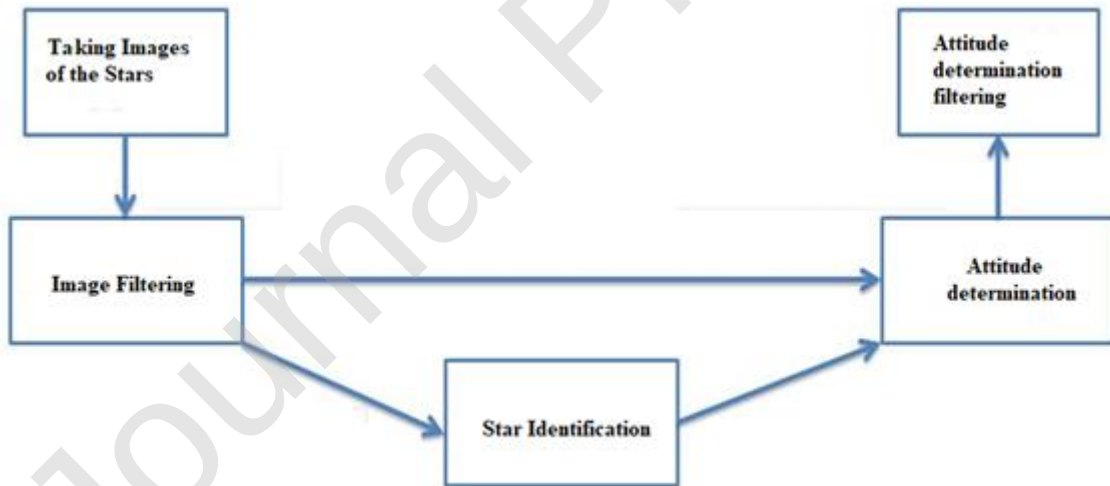


Fig.2. Attitude determination processes of a conventional star tracker.

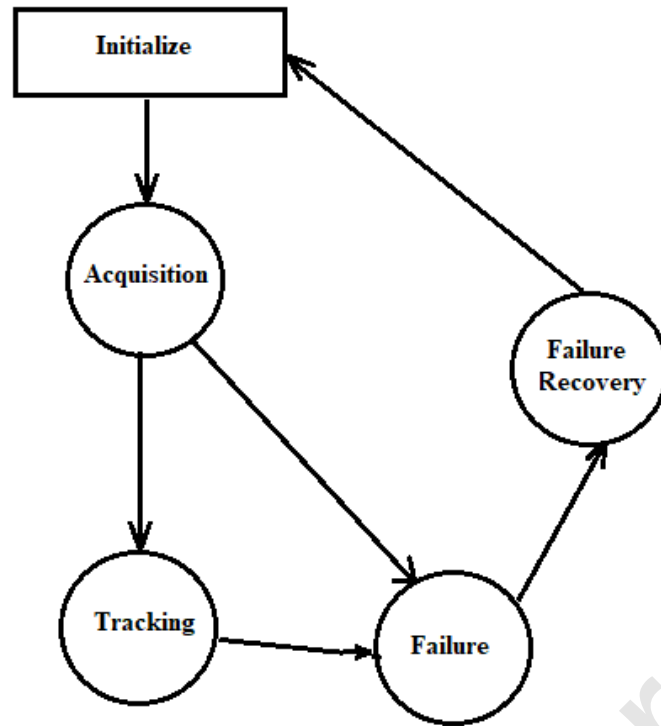


Fig. 3. Star tracker state diagram.

2-1- Image processing

The angular size of the Earth for a near-Earth satellite is quite large as compared to that of some of the stars and Sun treated as point source [15]. This implies that their images are inside the sensor's instantaneous field of view, but the accuracy of the positions of the instantaneous field of view is too weak to obtain a proper performance. As a result, sub-pixel accuracy is required for angular position calculations. Because of diffraction effects however, the image of each star is not a point of light but a disk. Therefore, the light emitting point sources, which are processed by the optical system, instead of being focused at a focal point, are spread on a number of vague points over the image.

When this point source is examined, the intensity distribution can be expressed by a mathematical function. This function is known as the point spread function (PSF) of the lens. For an object that is spread across a surface or several pixels, this object can no longer be considered as a point. Therefore, its coordinates or its location should be attributed properly. To identify the coordinates of an object, the center of the surface is selected as the representative of the location of that object. The center of the surface is estimated by the centroid of the object. In this sensor, star-noise detection algorithm based on the fixed-sampling window was used. A 3×3 window was used to calculate the star centroid.

2-2- Matching algorithms for identify stars in the image

In matching algorithms, firstly, stars are connected to create geometric shapes. Then, some geometric properties of these shapes are extracted and matched with the properties of the star catalog that have been extracted in the same way. In this star tracker inter-star angular separation was selected. As a result, the existing stars are identified in the image, and using the orientation of the created geometric shapes, the satellite orientation is determined.

2-3- Image feature extraction for pattern matching

With the choice or combination of some image features, the accuracy of the star tracker can be improved. These features include the star illumination, size (magnitude), number of stars, inter-star distance, inter-star angle, and star energy. Among the above features, a number of features or a combination of them should be selected. Different methods and algorithms can do choosing these features. Some important methods include:

- Comprehensive search method: All possible combinations are evaluated to provide the best combination.
- Sequential search method: In this method, an objective function is used as a criterion for evaluating and selecting features. This objective function can be selected according to the accuracy of the star tracker, correlation between the features, entropy of the features, and number of the features, or even a combination of them. Selecting an objective function is based on the most key goal (which is accuracy) or a combination of the required goals. In this sensor, inter-star angular separation based Pyramidal algorithm was selected as the main “lost in space” mode algorithm [16]. This algorithm was selected due to low computational complexity and robustness to calibration errors caused by changes on optical properties. Practical results show that this algorithm is capable of identifying more than 96% of the stars in the image.

2-4- Detecting stars in the taken image

Due to spherical aberration, the image is blurred; therefore, the energy is distributed over more than one pixel of our image detector [17]. This reduces the signal-to-noise ratio of the image and makes it difficult to distinguish the star from background noise. In addition, due to the sensitivity of different parts of the image sensor varies, choosing a constant threshold for star diagnosis will not lead to satisfactory results. For this reason, this star-tracker uses fixed sampling window method. The size of these windows has been determined by the optical characteristics and various field tests.

In order to extract the noise level and obtain the threshold for star detection, a grid is created by square reticulation of the images taken from the stars, where each square contains 35 pixels. In each grid, first the average illumination signal is calculated and considered as background noise. The images taken from stars look like Fig. 4. As seen, the image has a peak that is different from noise. The basis of the star detection algorithm is the distinction between the stars in the image and its noise. We use a 3×3 mask to recognize the single-star image pattern. Considering Fig. 5, we need three different thresholds with $T_1 > T_2 > T_3$ to detect the central

pixels and side pixels, where T_1 is the central pixel detection threshold, T_2 is the side pixel detection threshold, and T_3 is the diameter pixel detection threshold. The value of each threshold is determined as follows:

$$\begin{aligned} T_1 &= k_1 \times I_{avg} \\ T_2 &= k_2 \times I_{avg} \\ T_3 &= k_3 \times I_{avg} \end{aligned} \quad (10)$$

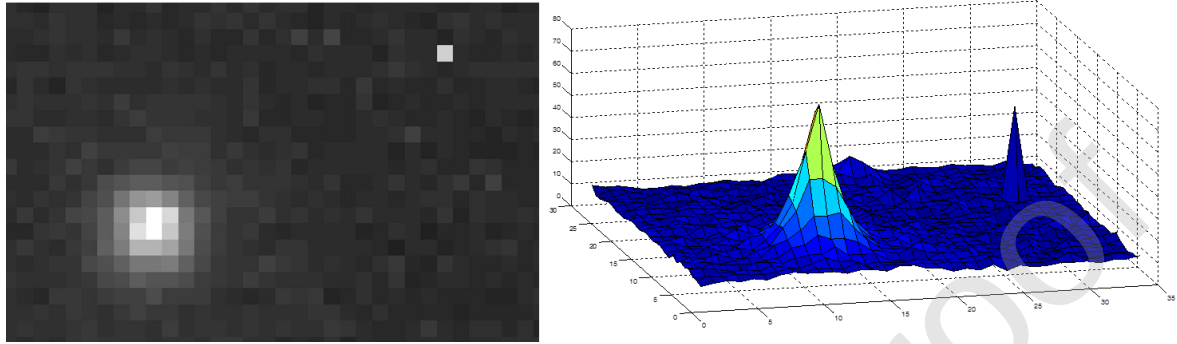


Fig. 4: Star image and its amplitude.

In order to detect a pattern as a star, in addition to the high amplitude of the signal relative to the threshold, the shape of the image should follow the pyramid pattern. Therefore, to identify the image, we should find the brightest pixel that is larger than the threshold of T_1 . After finding this pixel, if the neighbor pixels are also larger than the corresponding thresholds, and their amplitudes relative to the central pixel follow the forthcoming formula, it will be identified as a star:

$$\begin{aligned} P(x, y) &> P(x-1, y-1), \quad P(x, y) > P(x-1, y), \\ P(x, y) &> P(x, y-1), \quad P(x, y) > P(x+1, y+1), \\ P(x, y) &> P(x+1, y), \quad P(x, y) > P(x, y+1) \\ P(x, y) &> P(x+1, y-1), \quad P(x, y) > P(x-1, y+1) \end{aligned} \quad (11)$$

After star detection, a 5×5 mask is utilized to determine the star centroid; see Figs. 6 and 7.

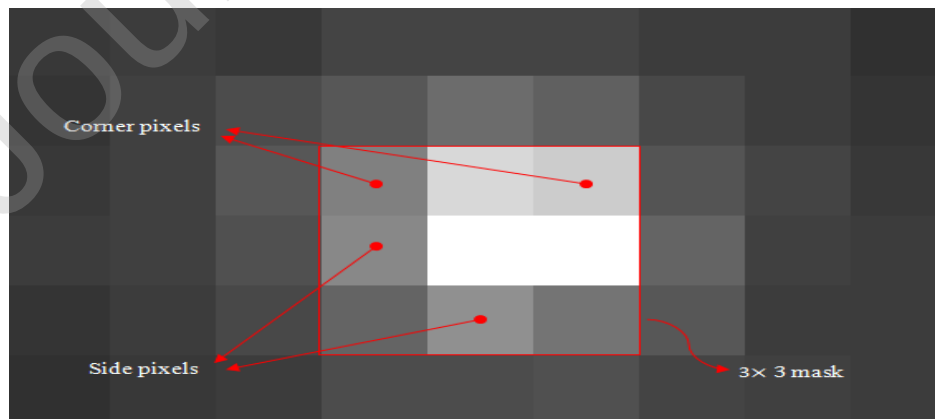


Fig. 5: Central pixel and side pixels.

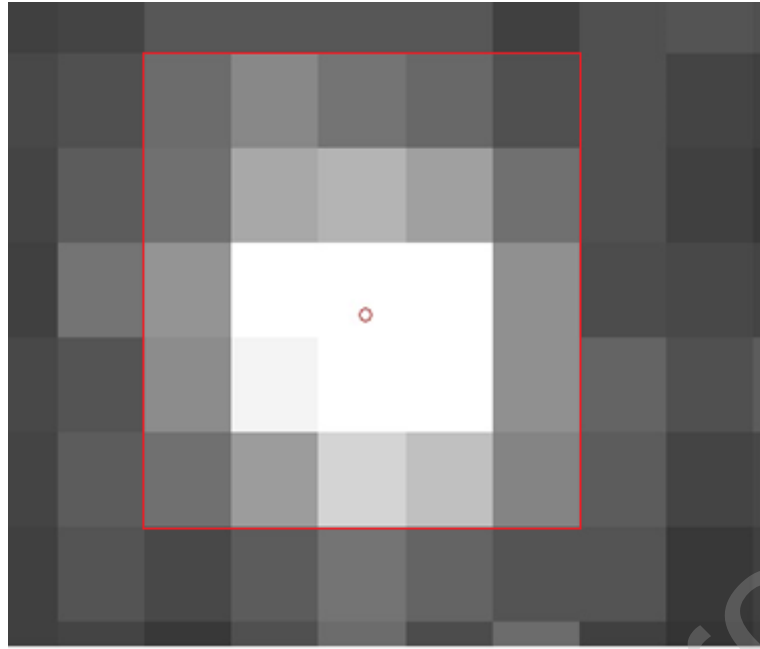


Fig. 6: star centroid.

2-5- Tracking algorithm

The star tracker works in two modes, namely initial positioning mode and tracking mode. Initial positioning mode is also called lost in space mode, and is applicable when the star tracker starts or when the system experiences a failure. In this mode, the pattern of stars in the field of view should be identified. In this mode, ensuring that stars are correctly identified is more important than the accuracy of measuring the position. In this mode, since no prior information is available, star identification in the whole sky is needed to find a primary position. Usually, identification is done in a few seconds. When this initial position is detected, the star tracker enters the tracking mode. Tracking mode tracks the previously identified stars in known positions.

There are many tracking algorithms/methods, which can be used by star trackers [18]-[23]. Unfortunately, limited functional information on star tracking algorithms has been published. Famous methods of identification and tracking algorithm include spherical polygon search [24], star neighborhood approach [24], star identification algorithm, and Zhang algorithm [16]. In the tracking algorithm used in this star tracker, without using the speed information, positioning data from previous image frames is used for tracking. Because of not using the satellite speed information and also utilizing common star catalog with the “lost in space” mode, star tracking owns high speed implementation, low cost, and low memory requirement. The tracking algorithm used in this tracker is as below [25]:

1. All neighbors for detected stars (pivot star) will be extracted from the catalog.
2. The stars which are able to be in the image are determined, according to attitude determination in lost in space mode (or previous tracking step) and optical geometry of the star

tracker. A window will be created surrounding each possible star. The dimensions of this window are relative to maximum angular velocity of the satellite and also relative to pixel to angle ratio.

3. A new image is captured from stars.

4. The new image is searched for a star in detecting window and center of the stars are obtained by using centroiding algorithm.

5. If the pivot star was also in the new image, the angles between this star and the other determined stars are calculated and if this calculated angle was in the measurement tolerance of the star tracker, the determined stars recognized as real star.

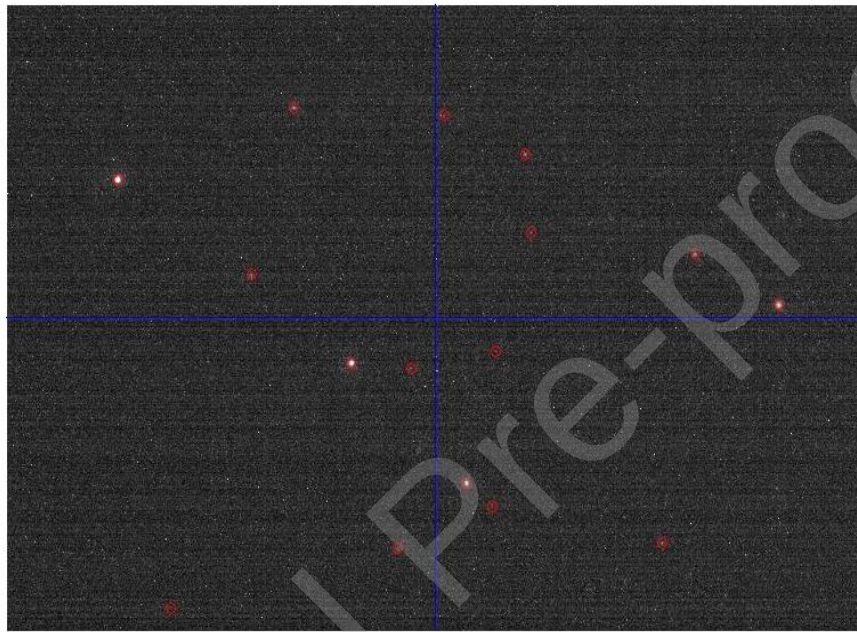


Fig.7: Identified stars in image.

3- Electronic hardware of the star tracker

Figure 8 shows the block diagram of the star tracker. The processor is responsible for executing software and algorithms for image processing and tracking as well as control of communication equipment and camera. Digital interface unit is responsible for receiving raw data from the camera and delivering this information to the processor. It also performs control of shooting timing, information retrieval and framing, and implementing serial and parallel communications. The bus controller establishes the hardware link of the star tracker with the external units. In this sensor, the RS485 bus is used to connect the star sensor to the control and determining position subsystem. In Fig. 9, the readout block diagram of the CMOS sensor is shown. The timing unit is responsible for regulating the registers and control signals and producing the required image sensor drivers. The digital processing unit also has the task of image processing, star identification, and tracking.

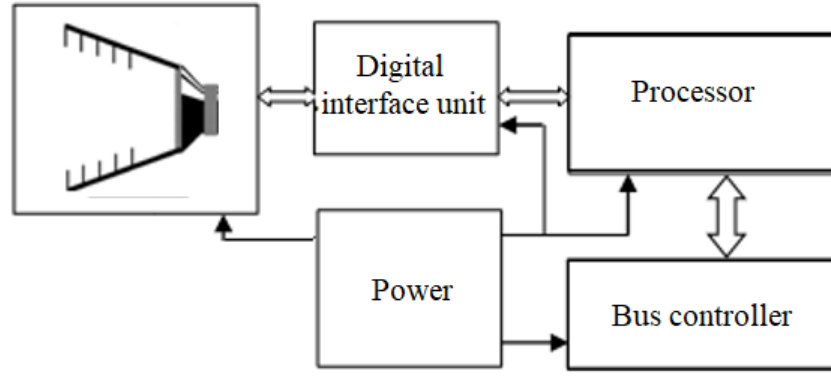


Fig. 8. Hardware block diagram.

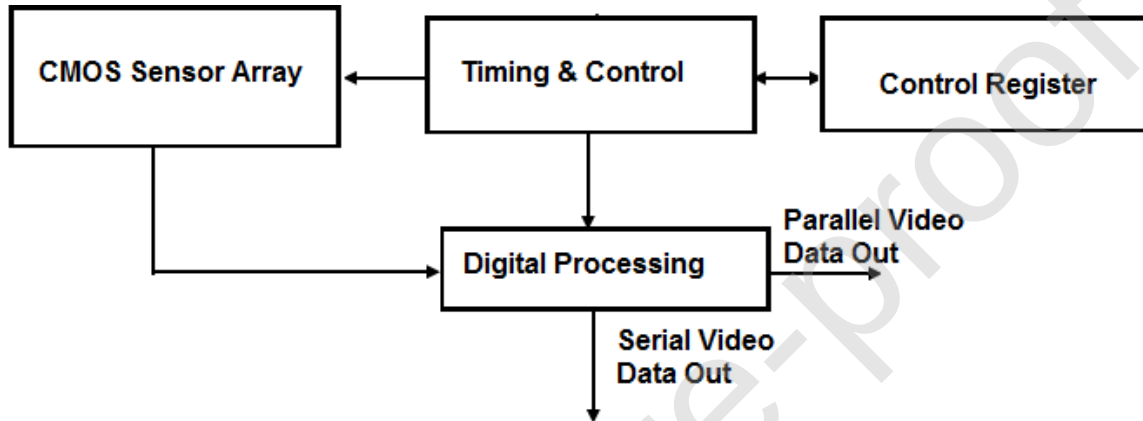


Fig. 9. Star sensor readout block diagram.

3-1- Image sensor

In the star tracker camera, various sensors can be used for imaging, the most important of which are CCD and CMOS sensors. Each of these sensors has its own advantages and disadvantages, which affect the star tracker image quality and physical characteristics such as the dimensions of the lens and field of view. Image sensors vary in terms of characteristics such as power consumption, image quality, speed performance, cosmic ray resistance, light sensitivity and physical dimensions. The sensor should be selected by considering the optical parameters and some other factors such as power limitation, required quality of images, star tracker update speed, field of view.

The main advantage of CMOS imaging systems is being adaptable to the silicon chip technology. Because of transistors built with this technology, intra-chip processing is possible. By aligning the processor on the chip, the CMOS sensor is physically smaller than the CCD, because the CCD sensor requires a separate hardware for the clocks, image formatting, and signal processing [25]. The other advantage of the CMOS is its lower power consumption than the CCD. In addition, CMOS sensors have higher resistance to cosmic radiations than the CCD sensors, and therefore have longer lifetimes.

The electron multiplying CCD (EMCCD) has been developed in recent years. Due to its high signal-to-noise ratio, simple structure and high dynamic, it has been applied in many aerospace

missions, such as ESO telescope and Planck telescope. The main features of three kinds of detectors applied in star trackers are shown in Table 1 [26].

The CMOS sensor has been used in satellites such as ITU-PSAT-1, AAUSAT, COMPASS-1, Cubesat, CAN-X1 [27]. That is why this technology has been used to create a star tracker. We used MT9P031 CMOS sensor with a resolution of 5Mp, which is produced in both colored and black and white types. The Aptina® MT9P031 is a 1/2.5-inch CMOS active pixel digital image sensor with an active imaging pixel array of 2592H x 1944V. The sensor has a high dynamic range and a signal-to-noise ratio of 64dB, a maximum clock frequency of 148.5 MHz, and a frame rate of 14 frames per second. The detector uses 752(H) × 480(V) pixels of MT9P031 sensor. One of the features of this sensor is the possibility of working in low light conditions and with a slight dark flow. Due to the use of DigitalClarity® technology, CMOS provides a sensor with a high level of CCD sensor while working at low power levels.

This sensor is attractive because of its very low readout noise, but achieving consistent high quality image over lifetime of the device presents some challenges [28]. It incorporates sophisticated camera functions on-chip such as windowing, column and row skip mode, and snapshot mode. It is programmable through a simple two-wire serial interface. This sensor can also set the analog-to-digital gain, correct the black level, compensate the offset, and control the exposure time. Processor writes the read control into the sensor via I2C [29], [30]. It can also read out a region of interest (ROI) instead of the whole sensor, for increased frame rate, and it has integrated black-level correction and analog gain. Also supported is a global-reset exposure mode, as well as the typical rolling-shutter mode—in global-reset mode, a mechanical shutter is needed to guarantee equal exposure times for all pixels. It can also operate in snapshot mode, where each exposure is triggered either through a register write or through an external signal. All the registers can be accessed by the two-wire serial interface with 16-bit addresses and 16-bit data. Accessing the sensor registers is done over the standard I2C bus, operating at 400 kHz [30].

This image sensor is one of the most popular sensors in the field of star tracker, which has been used by different countries. The sensor is used in the ST-16 star tracker, which is a fully automatic tracker with a weight of 90 grams (without baffle), usage power of less than 1 Watt, and an arched accuracy of 7 arc seconds. This star tracker has been produced in 17 models and put in orbit since 2013 [31]. This sensor is also used in the ST-200 star tracker whose processor is ARM9 [36]. The MT9P031 sensor in NASA and Canada has been tested under radiation, and the results indicated the success of the sensor for work on LEO missions. The MT9P031 sensor was tested in 2010 at the TRIUMF facility in Vancouver, Canada, which is equipped with 105 MeV proton probes. The test results indicated that this sensor could be used in LEO missions [28].

MT9P031 supported has a global-reset exposure mode, as well as the typical rolling-shutter mode. In global-reset mode, a mechanical shutter is needed to guarantee equal exposure times for all pixels [32]. Mechanical shutters are not used, to keep the design simple and avoid the risk of damage to mechanically moving parts during the launch. Therefore, we have used rolling shutter mode in this star tracker. Finally, the pin-out of the image sensor MT9P031 is shown in Fig. 10.

3-2- Processor

The processor performs noise reduction and implements different algorithms including image processing and tracking algorithms. This star tracker uses a Texas Instrument OMAP3530 as processor. This processor supports powerful graphics interfaces, which reduces the size and power consumption of the star-trackers. The Open Multimedia Applications Platform (OMAP) is a small sized processor with powerful interfaces and high processing capabilities. The processor has the following serial channels: UART1, UART2, UART3, one SPI Ports and I2C ports. Some features that are implemented on the OMAP3530 include real time clock (RTC), Secure Digital, graphics accelerator, DSP codecs, Image Processing Unit. The hardware supports high-level operating systems such as Linux, Windows CE, Symbian OS and Palm OS. Moreover, the processor has a power management system that reduces power consumption. This processor has been used as the main processor of the star tracker of st-16 (with 17 successful launches) [33]. In addition, this processor has been used as the command and data handling (C & DH) system in Trailblazer-2 and CubeSat satellites. The processor has been tested during 2012 and 2013 using cobalt 60 at a rate of 10rad (Si) / s. All components of the electronic board have been in operation. The testing process consisted of memory, input / output and communications ports like USB, HDMI and AUDIO. Test results illustrated capability of operating of the board up to 35 krad (Si) without requiring any shielding [33].

Figure 11 shows the processor block diagram and its interfaces. The star-tracker is directly connected to the onboard computer of the satellite using CAN bus. This lets star-tracker receive its latest attitude data using OBC in the case that ADCS has encountered a fault. In addition, the settings for the MT9P031 sensor are performed via the I2C bus and the registers of this sensor are set. The MT9P031 sensor has 256 registers whose setup determines the sensor status. The setting of these registers is based on the mission protocol and is performed by the star tracker processor via the I2C bus. I2C transmits data through two signal wires, one for serial data and another for the serial clock. Therefore, the initialization and configuration of the MT9P031 sensor are performed via the I2C bus. The ARM microcontroller of the processor generates other required timing signals and drivers for this sensor. The memory used in this star tracker is SDRAM, NOR Flash, and NAND Flash. Considering that the NOR flash memory has low write speed and high read speed, these memories are used to store the main program. The part number of the used NOR Flash and NAND Flash is S25fL127sabmfb000 and

Mt29f4g01adagdw-b it respectively. In addition, similar write and read speeds of the NAND Flash memory, they are used to store data. The output data of the star tracker is transmitted via the RS485 bus or directly to the attitude determination and control subsystem. To adjust and control CAN serial bus, a SAL-XC886CLM-8FFA 5V AC microcontroller has been utilized. The micro controller that is used as a supervisor to the OMAP3530. The SAL-XC886CLM processor turns the OMAP processor on for the image acquisition and attitude determination phases. Moreover, UART provides an interface for communicating with the user during scheduling and testing. Figure 12 shows the schematic of the processor. The fabricated star tracker is shown in Fig. 13.

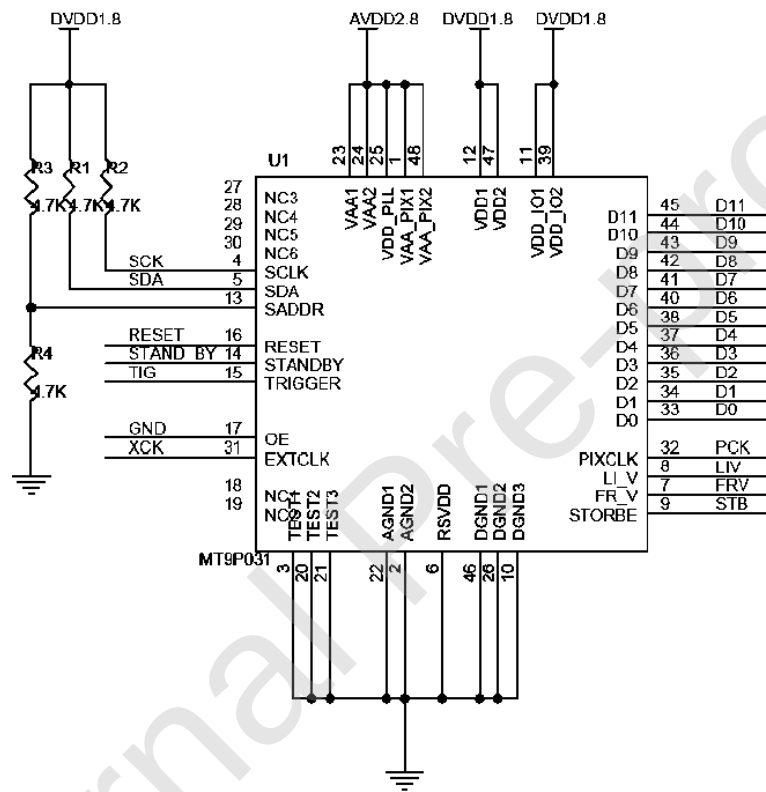


Fig. 10. The schematic of MT9P031 circuit

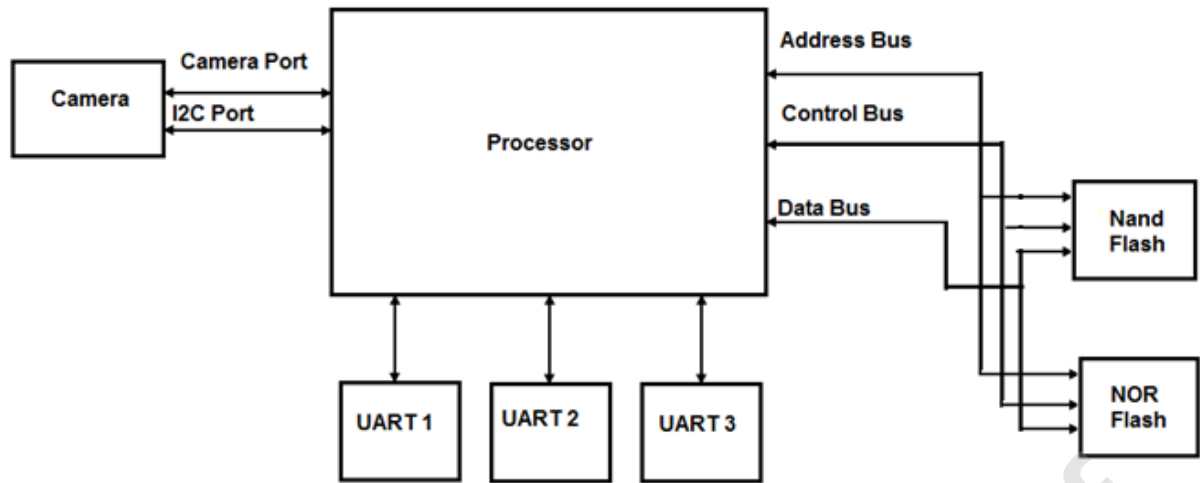


Fig.11. Block diagram of star tracker processor

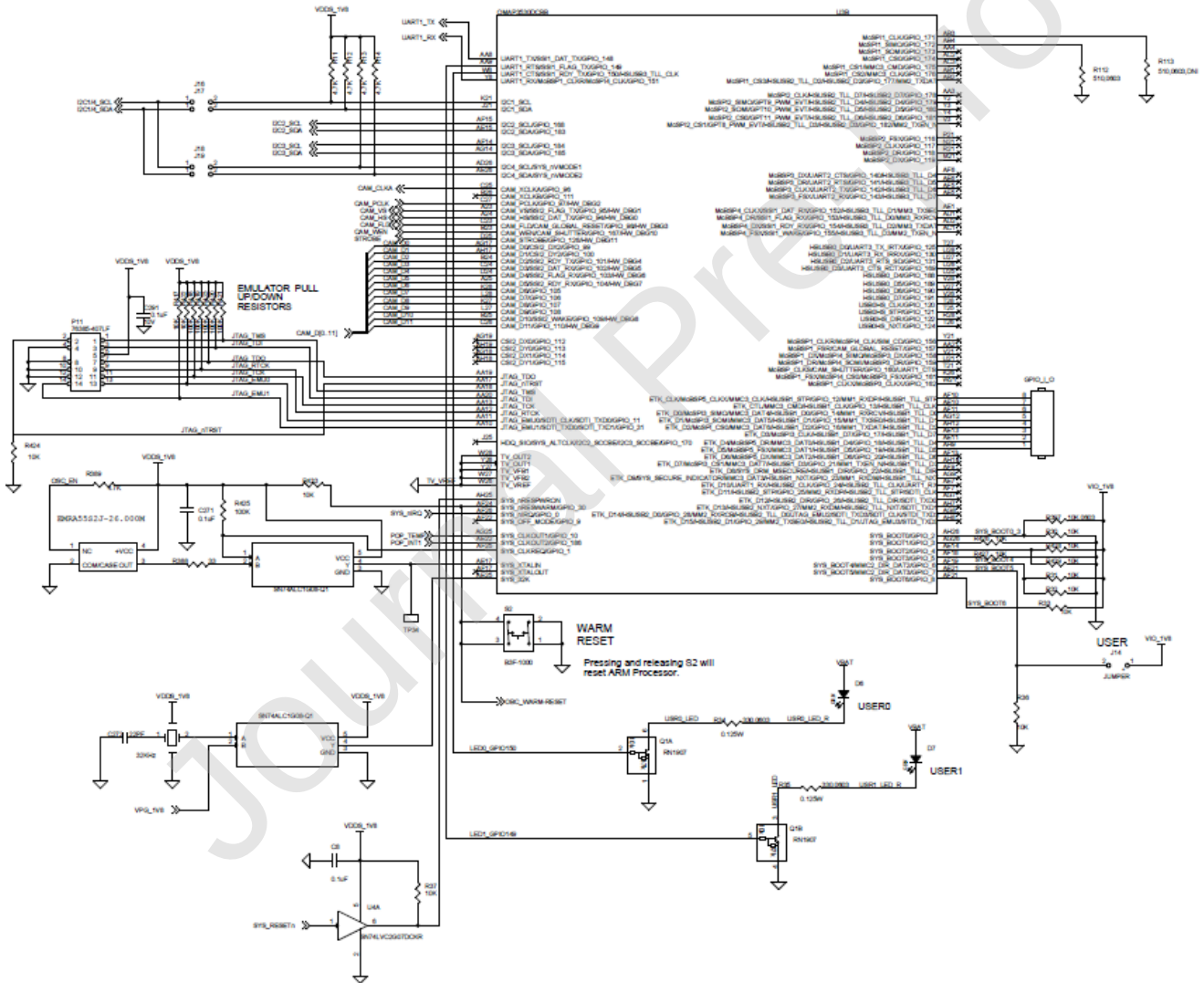


Fig.12. Processor and its interfaces.



Fig. 13. The fabricated star tracker

4- Test results

The purpose of this test was evaluating the ability of the tracker to correctly identify the stars. For this purpose, numerous images were taken either continuously or discontinuously by the tracker, and the tracker's ability to identify the stars in the image was evaluated using the stellarium software. The system succeeded in identifying stars with a magnitude of 7.2. From the photographs taken at different times and on different stars, three images were not identified and no false identification was observed during the test. From the three unsuccessful images, one lacked enough stars and the two other images could not be identified due to being on the horizon and the effect of atmospheric distortions. According to the tests, the success rate was 96.81%. Note that because there is no effect of atmospheric distortions in the satellite orbit, this accuracy will be improved. In Table 2, a comparison was made between our constructed star sensor and some of the existing sensors. As seen, the parameters obtained for our star tracker is satisfyingly comparable to the operational ones made in the world.

5- Conclusion

In this paper, the design and implementation of a star tracker was considered for a LEO satellite. The hardware used and various algorithms of this sensor are presented and the results of practical tests are mentioned. The sensor is CMOS-based MT9P031 and the processor is OMAP3530. Different algorithms for image processing and tracking were presented. Inter-star angular separation based pyramidal algorithm was selected as the main “lost in space” mode algorithm. The test results were satisfying with regard to the effects of atmospheric distortion at the apparent location of the stars on the sky and the effect of this displacement at inter-star

angles. During the test, no mistake identification were performed and the identification success rate of this system was 96.81%.

References

- [1] Kandiyil, R.: Attitude Determination Software for a Star Sensor, Lulea University of Technology, Germany, 2009.
- [2] Magnitude System, March 2019, <https://www.astronomynotes.com/starprop/s4.htm>
- [3] E. Silani and M. Lovera, "Star Identification Algorithms: Novel Approach & Comparison Study", *IEEE Trans. on Aerospace and Electronics Systems*, Vol. 42, 2006, pp. 1275-1287.
- [4] C. G. Smith, "Development and implementation of star tracker based attitude determination," *tech. rep.*, Missouri, 2017.
- [5] John Leif Joergensen and Andrew J. Pickles "Fast and robust pointing and tracking using a second-generation star tracker", *Proc. SPIE 3351, Telescope Control Systems III*, (26 May 1998);
- [6] M. Na, P. Jia, "A survey of all-sky autonomous star identification algorithms", 2006 *1st International Symposium on Systems and Control in Aerospace and Astronautics*, Harbin, China, pp. 896–901. IEEE, Los Alamitos (2006).
- [7] Malak 3, Stephan Theil, "Development of a low cost star tracker for the SHEFEX mission", *Aerospace Science and Technology*, Vol. 23, Issue 1, pp. 469-478, December 2012.
- [8] Uwe Schmidt, "ASTRO APS - The Next Generation Hi-Rel Star Tracker Based on Active Pixel Sensor Technology", *AIAA Guidance, Navigation, and Control Conference and Exhibit, Guidance, Navigation, and Control and Co-located Conferences*, 2005, DOI: 10.2514/6.2005-5925.
- [9] U. Schmidt. FikselA. Kwiatkowski. Steinbach. Pradarutti. Michel. Benzi, "Autonomous star sensor ASTRO APS: flight experience on Alphasat ", *CEAS Space Journal* June 2015, Volume 7, Issue 2, pp 237–246.
- [10]. Meiying 6, Hu Wang, Desheng Wen, Shaodong Yang, "A Novel Design of Subminiature Star Sensor's Imaging System Based on TMS320DM3730", *Proc. of SPIE*, Vol. 10256 1025635-1.
- [11] Alexander O. Erlank, Willem H. Steyn., "Arcminute Attitude Estimation for CubeSats with a Novel Nano Star Tracker ", *Proceedings of the 19th World Congress the International Federation of Automatic Control*, Cape Town, South Africa. August 24-29, 2014.
- [12] Zunzong Ye, Xinguo Wei and Gangyi Wang, "Design of Star Sensor Imaging Driver Module based on IMX224 ", *Advances in Materials, Machinery, Electronics I AIP Conf. Proc.* 1820, 070004-1–070004-7; doi: 10.1063/1.4977346
<https://pdfs.semanticscholar.org/7db7/c9a2c60ee9804d1a793da20e38bb24d2504b.pdf>
- [13] Liebe, C. C. "Accuracy performance of star trackers - a tutorial ", *IEEE Transactions on Aerospace and Electronic Systems*, vol. 38, issue 2, 2002.

- [14] Benjamin B. Spratling, IV and Daniele Mortari, "A survey on Star Identification Algorithms", *Algorithms*, 2009 volume 2, 93 – 107.
- [15] Lecno, Attitude determination sensors, lecture note, <https://nptel.ac.in/courses/101106046/Lecture26.pdf>
- [16] Mortari D, Junkins, J L., 2001. "Lost-in-space pyramid algorithm for robust star pattern recognition", 24th Annual AAS Guidance and Control Conference. Breckenridge CO, AAS Paper 01-004.
- [17] Birkfellner, *Applied Medical Image Processing: A Basic Course*, CRC press, 2014
- [18] J. Jiang, G. Jun Zhang, X. Wei & X. Li, "Rapid Star Tracking Algorithm for Star Sensor", *IEEE A&E Systems Magazine*, September 2009, pp. 23-33.
- [19] Accardo, D. and Rufino, G., "Brightness-independent star-up routine for star trackers", *IEEE Trans. on Aerospace and Electronic Systems*, Vol. 38, 2002, pp. 813-823.
- [20] Baohua U, Yingchun Zhang, Huayi Li, etc., "A star tracking algorithm suitable for star sensor", *Proc. SPIE* 6595, 659523, (2007).
- [21] J. Jie, Li X., Z. Guangjun, etc., "Fast star tracking technology in star sensor", *Proc. SPIE* 6358, 63580A, (2006).
- [22] Alexander J. and Chang, D., "Star tracking and identification algorithm, scene simulation, and testing", *SPIE*, Vol. 2803, 1996.
- [23] Quine, B.M. and Durrant-Whyte, H.F., "Rapid star pattern identification", *SPIE*, Vol. 2739, 1996.
- [24] C Padgett, K Kreutz-Delgado, "A grid algorithm for autonomous star identification", *IEEE Transactions on Aerospace and Electronic systems*, 33 (1), 202-213, 1997.
- [25] Holst GC, *CCD Arrays, Cameras, and Displays*, 2nd ed.. JCD Publishing & SPIE Press, 1998.
- [26] Mehdi Nasiri-Sarvi, Shahram Mohammadnejad, HeshmatAllah Hosseini Ghiasvand, "A Novel Gyro-less Tracking Method for Star Tracker" *International Journal of Engineering Sciences & Research Technology (IJESRT)*, Vol.3, No. 4, 2014.
- [27] Penghai Wang, Lidong Lan, Yifei Han, Guanya Wang, and Haiyang Quan, "Design of a Miniature CMOS APS Star Tracker", *International Journal of Electronics and Electrical Engineering*, Vol. 4, No. 1, February 2016.
- [28] K. Khurshid, R. Mahmood, Q. ulIslam, "A Survey of Camera Modules for CubeSats - Design of Imaging Payload of ICUBE-1", *6th International Conference on Recent Advances in Space Technologies (RAST)*, 12-14 June 2013, Istanbul, Turkey.
- [29] Enright, John, Doug Sinclair, K. Christy Fernando, "COTS Detectors for Nanosatellite Star Trackers: A Case Study", *25th Annual AIAA/USU Conference on Small Satellites*, Logan, Utah, 2011.
- [30] K. Grabowski, J. Clapa, "A Low-Power Embedded System for Visual Vehicle Detection", *MIXDES 2014, 21st International Conference "Mixed Design of Integrated Circuits and Systems*, June 19-21, 2014, Lublin, Poland.

- [31] Eino-Ville Talvala, The Franken camera: Building a Programmable Camera for Computational Photography, Ph.D. thesis, Stanford University, Feb. 2011.
- [32] Small Spacecraft Technology State of the Art – 2014, https://www.nasa.gov/sites/default/files/atoms/files/small_spacecraft_technology_state_of_the_art_2014.pdf
- [33] Fernando, K. C., “Hot Pixel Detection for Star Trackers,” M. S., Ryerson. University, 2009.

Table 1. Features of CCD, CMOS and EMCCD

Detector	sensitivity	Complexity	Sampling rate	power	sensitivity
CCD	High	Complex	Low	High	High
CMOS	General	Simple	High	Low	General
EMCCD	Ext-high	Complex	Low	Ext-high	Ext-high

Table 2. Comparison of our fabricated star sensor and existing ones.

no	Name	Manufacturer	Accuracy (arcsec)	Wight (kg)	Power (watt)	Update (Hz)	FOV (deg)	Star Mag. (Mv)	Rate (deg/sec)
1	ALTAIR HB+	Surrey	10	2.6	12	1 (max)	10×10	-	0.2 deg/s
2	CT-601	Ball Aerospace & Technologies Corp	3	7.8	8~12	10	8×8	1~6	0.3
3	VST-41M	VECTRONIC Aerospace	18	1.1	2.5	4	14×14	-	-
4	Star Tracker	AXELSPACE	30	-	1.8	1	8×8	-	-
5	Star Tracker	satellite services Ltd	22	2.61	<3	1	18.6×12	6	0.5
6	SED36	EADS SODERN	17	-	7	8		-	-
7	HE 5AS	TERMA	5	-	6.8	4 Max		-	0.5
8	star tracker	SUN SPACE	15	-	<3	1	18.6×12	6	0.5
9	fabricated Star Tracker		24	2/5	<2.5	4	8×8	2.5~6.5	0.5